

Brain Cancer Machine Learning Classification

COSC 525

GRANT ALDERSON, UTKARSH KARKI, and DAVID MUSILI



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

Problem Definition

Brain tumor Classification is a critical task in modern medical diagnostics.

This classification is typically a time consuming process that can be automated using machine learning.

With that in mind we created several models that successfully automate this task using machine learning architectures.

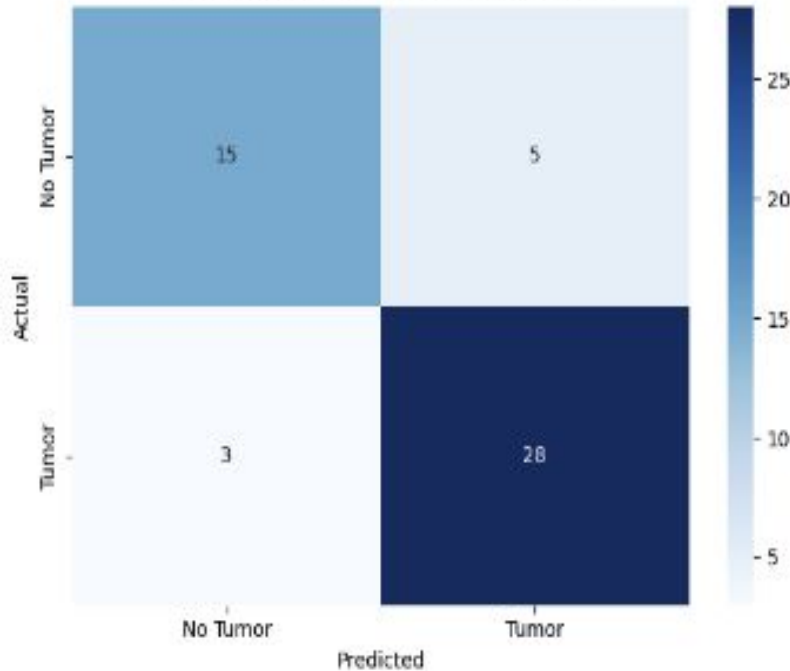
Dataset and Encoding Strategy

The Dataset used for this project contains MRI images of brains with and without tumors. The Dataset then further categorises the Images with a tumor into 3 different types.

The MRI images were then label encoded and split into training and testing sets for evaluation.

Base Logistic Regression

Confusion Matrix — boolcancer Test Set



This regression model is the baseline model for the Yes/No predictions.

It uses 64x64 grayscale rawpixel values that have been flattened and normalized and uses a simple logistic regression architecture.

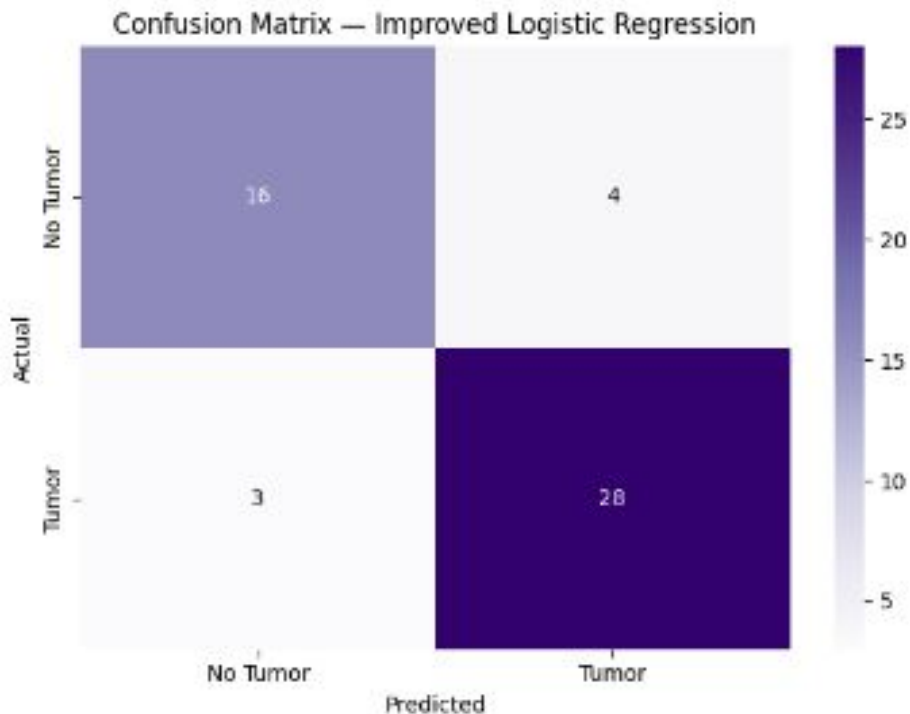
Accuracy : 0.8431

Precision: 0.8485

Recall : 0.9032

F1-Score : 0.8750

Improved Logistic Regression



The improved model uses logistic regression within a pipeline and improves the model by using standard scaler for feature scaling and class weight balance to address dataset imbalance.

Accuracy : 0.8627

Precision: 0.8750

Recall : 0.9032

F1-Score : 0.8889

CNN

The CNN model was trained specifically to predict the tumor type and then uses the confidence score for binary classification.

The CNN also uses a tensorflow keras architecture which includes a dropout layer and group normalization and several other optimizations.

Binary accuracy: .568

Multi classification accuracy: .98

ResNet18 + Logistic Regression

This model uses a pretrained ResNet 18 model as a feature extractor followed by a logistic regression model.

Uses 512 dimensional deep feature extraction.

Binary accuracy: .803

Fine-Tuned ResNet18 CNN (Binary)

This model uses a pretrained ResNet18 model as a feature extractor and a CNN.

The training works by first training only the new classification head using frozen CNN layers and then unfreezing all the layers and training with a smaller learning rate.

It was also further improved using dropout, class weighting, Data augmentation and early stopping.

Binary accuracy: .921

4-Class ResNet18

This model was designed to classify based on if it is a tumor or not as well as the type.

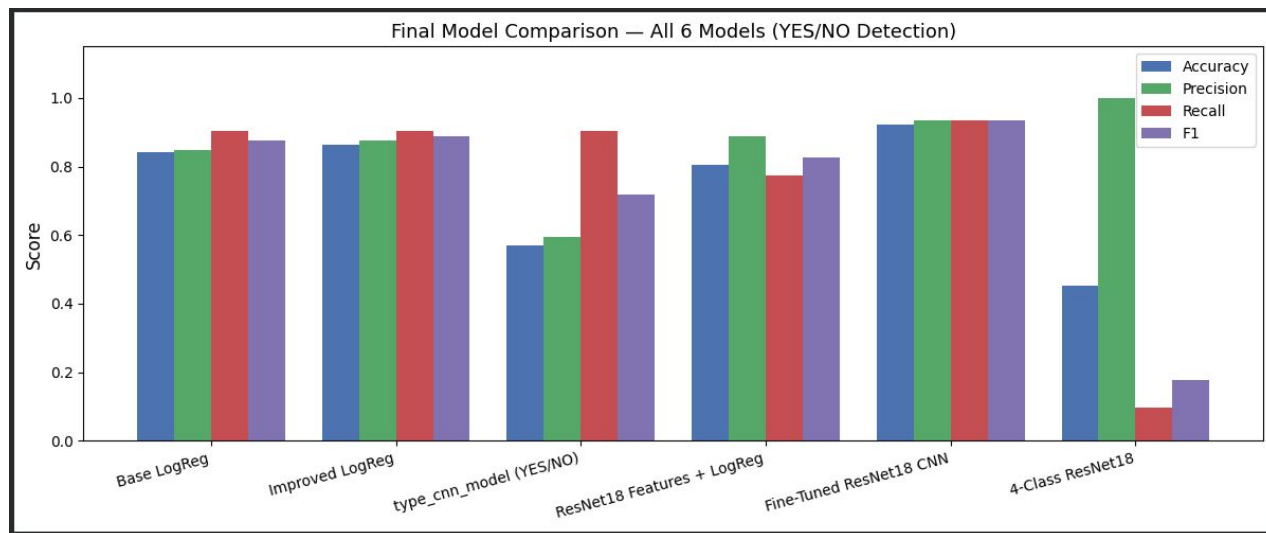
Output classifications: 'No Tumor', 'brain_glioma', 'brain_menin', 'brain_tumor'.

Similar two-phase fine-tuning as previous model, with class weighting for imbalance.

Binary accuracy: .451

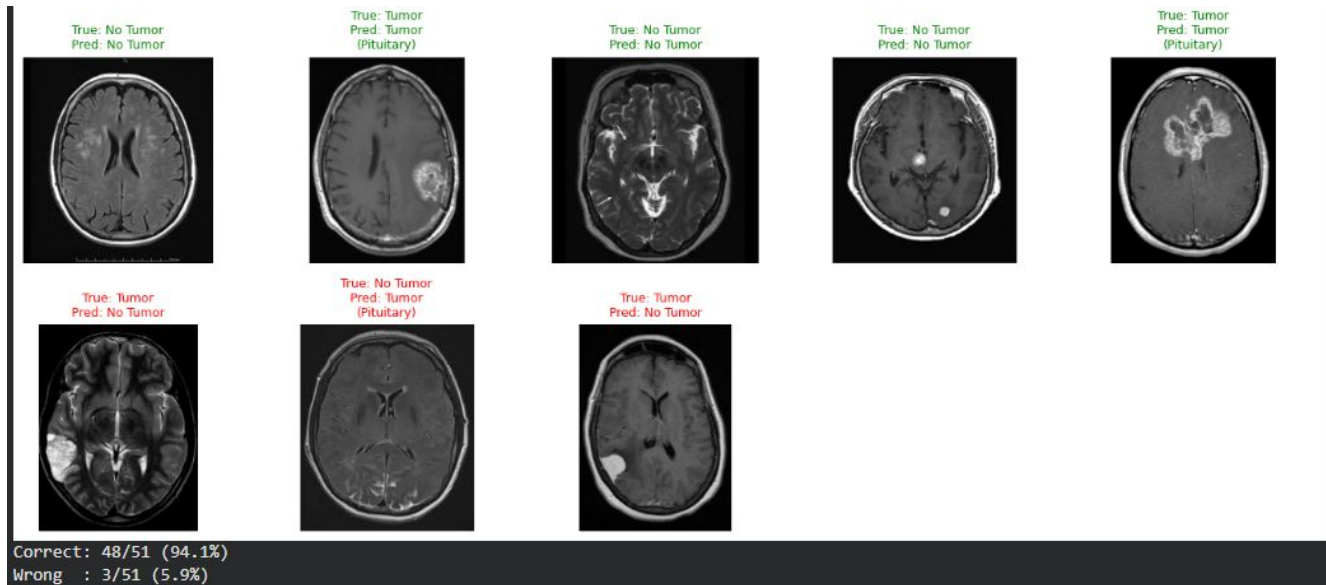
Results

The ResNet18 CNN performed best for the binary classification of the MRI images.



Results

Example results
from our best
binary
classification
model Fine-tuned
ResNet 18 CNN



Conclusion

The logistic regression models provided solid starting point for the model with an accuracy around .84-.86 with slight improvements from the standard scaler and class weighting.

The CNN for tumor classification had a high accuracy of .98 when classifying between the types of tumors but its binary classification based on a confidence interval overall failed.

The ResNet18 CNN had the highest accuracy for binary classification at .92. This showcases the power of transfer learning for feature engineering.